Supplementary Material for Spanning the Narrative Space: State Transition Matrix Matching for Multi-Shot Video Evaluation

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1 Completeness Theorem: Full Proof

Theorem 1.1 (Completeness of the Narrative Space). *Under column permutation $\mathcal{P}_{ij}$ and integer-coefficient linear combination, the two basis matrices $\mathcal{B}_1$ (Relay) and $\mathcal{B}_2$ (Split) span $\mathbb{M}_{2\times3}(\mathbb{Z})$.*

Proof. We construct all six standard basis matrices E_{ij} from $\mathcal{B}_1$ and $\mathcal{B}_2$.

Step 1: Column isolation. Define three permutation variants of $\mathcal{B}_1$: $v_1 = \mathcal{B}_1 = E_{12} - E_{21}$, $v_2 = \mathcal{P}_{13}(\mathcal{B}_1) = E_{12} - E_{23}$, $v_3 = \mathcal{P}_{23}(\mathcal{P}_{12}(\mathcal{B}_1)) = E_{11} - E_{23}$. Computing $X = v_3 - v_2 + v_1$:

$$X = (E_{11} - E_{23}) - (E_{12} - E_{23}) + (E_{12} - E_{21}) = E_{11} - E_{21}. \quad (1)$$

Step 2: Dimension breaking via $\mathcal{B}_2$. Apply column permutation: $\mathcal{P}_{12}(\mathcal{B}_2) = -E_{11} + E_{21} - E_{22}$. Adding to X:

$$X + \mathcal{P}_{12}(\mathcal{B}_2) = (E_{11} - E_{21}) + (-E_{11} + E_{21} - E_{22}) = -E_{22}. \quad (2)$$

Thus $E_{22} = -(X + \mathcal{P}_{12}(\mathcal{B}_2))$.

Step 3: Propagation. From E_{22}, column permutations yield $E_{21} = \mathcal{P}_{12}(E_{22})$ and $E_{23} = \mathcal{P}_{23}(E_{22})$. From $X = E_{11} - E_{21}$ and the now-known E_{21}: $E_{11} = X + E_{21}$. Similarly, $E_{12} = \mathcal{P}_{12}(E_{11})$ and $E_{13} = \mathcal{P}_{13}(E_{11})$.

All six standard basis matrices $\{E_{11}, E_{12}, E_{13}, E_{21}, E_{22}, E_{23}\}$ are obtained. Any $\Delta S \in \mathbb{M}_{2\times3}(\mathbb{Z})$ is a unique integer combination of these. ☐

2 Topological Generalization to $\mathbb{M}_{(T-1)\times N}(\mathbb{Z})$

In Section 4, Theorem 1 proved that the basis set $\mathcal{B} = \{\mathcal{B}_1, \mathcal{B}_2\}$ spans the local narrative space $\mathbb{M}_{2\times3}(\mathbb{Z})$ under the symmetric group of column permutations Σ_3. Here, we prove that this local spanning property generalizes to the global multi-shot space $\mathbb{M}_{(T-1)\times N}(\mathbb{Z})$ for any $T \geq 3$ and $N \geq 3$.

2.1 Operator Definitions and Validity

To formalize the dimensional expansion, we define two linear operators.

Spatial Zero-Padding Operator ($\mathcal{Z}_N$). Let $\mathcal{Z}_N : \mathbb{M}_{2\times3} \to \mathbb{M}_{2\times N}$ be defined by appending $N-3$ zero-columns:

$$\mathcal{Z}_N(X) = \begin{bmatrix} X & 0_{2\times(N-3)} \end{bmatrix}. \quad (3)$$

Linearity. For any $c_1, c_2 \in \mathbb{Z}$ and $X, Y \in \mathbb{M}_{2\times3}$:

$$\mathcal{Z}_N(c_1X + c_2Y) = \begin{bmatrix} c_1X + c_2Y & 0 \end{bmatrix} = c_1\,\mathcal{Z}_N(X) + c_2\,\mathcal{Z}_N(Y). \quad (4)$$

Thus $\mathcal{Z}_N$ is a linear homomorphism preserving spanning properties. We define the padded bases $\hat{\mathcal{B}}_1 = \mathcal{Z}_N(\mathcal{B}_1)$ and $\hat{\mathcal{B}}_2 = \mathcal{Z}_N(\mathcal{B}_2)$.

Temporal Shift Operator ($\mathcal{S}_\tau$). Let $\mathcal{S}_\tau : \mathbb{M}_{2\times N} \to \mathbb{M}_{(T-1)\times N}$ be a block matrix insertion operator for temporal window $\tau \in \{1, \ldots, T-2\}$:

$$\mathcal{S}_\tau(X) = \begin{bmatrix} 0_{(\tau-1)\times N} \\ X \\ 0_{(T-\tau-2)\times N} \end{bmatrix}. \quad (5)$$

Linearity. For any $c_1, c_2 \in \mathbb{Z}$ and $X, Y \in \mathbb{M}_{2\times N}$:

$$\mathcal{S}_\tau(c_1X + c_2Y) = c_1\,\mathcal{S}_\tau(X) + c_2\,\mathcal{S}_\tau(Y). \quad (6)$$

The linearity of $\mathcal{S}_\tau$ guarantees that superposition of local transitions maps to the global space without cross-temporal interference.

2.2 Lemma 1: Spatial Extension

Lemma 1. *The set $\hat{\mathcal{B}} = \{\hat{\mathcal{B}}_1, \hat{\mathcal{B}}_2\}$ spans $\mathbb{M}_{2\times N}(\mathbb{Z})$ under the permutation group Σ_N.*

Proof. Let $E_{i,j}^{2\times3}$ be the elementary matrix in $\mathbb{M}_{2\times3}(\mathbb{Z})$. By Theorem 1:

$$\forall\, i \in \{1,2\},\ j \in \{1,2,3\} : \quad E_{i,j}^{2\times3} \in \operatorname{span}(\mathcal{B})^{\Sigma_3}.$$

Let $\hat{E}_{i,j}^{2\times N} = \mathcal{Z}_N(E_{i,j}^{2\times3})$. Since $\mathcal{Z}_N$ is linear, $\hat{E}_{i,j}^{2\times N} \in \operatorname{span}(\hat{\mathcal{B}})^{\Sigma_3}$.

For any target column $k \in \{1, \ldots, N\}$, there exists a permutation $\sigma \in \Sigma_N$ such that:

$$E_{i,k}^{2\times N} = \sigma\big(\hat{E}_{i,1}^{2\times N}\big).$$

Since any $\Delta S^{2\times N} \in \mathbb{M}_{2\times N}(\mathbb{Z})$ is a linear combination:

$$\Delta S^{2\times N} = \sum_{i=1}^{2}\sum_{k=1}^{N} w_{i,k}\, E_{i,k}^{2\times N}, \quad w_{i,k} \in \mathbb{Z},$$

it follows that $\Delta S^{2\times N} \in \operatorname{span}(\hat{\mathcal{B}})^{\Sigma_N}$. ☐

2.3 Lemma 2: Temporal Extension

Lemma 2. *The local basis spans the global temporal space $\mathbb{M}_{(T-1)\times N}(\mathbb{Z})$.*

Proof. Any $\Delta S \in \mathbb{M}_{(T-1)\times N}(\mathbb{Z})$ can be decomposed into $T-2$ overlapping local transitions. Let $\Delta S_{2\times N}^{(\tau)}$ denote the $2\times N$ submatrix of rows τ and $\tau+1$. The global matrix is the linear sum of shifted local blocks:

$$\Delta S = \sum_{\tau=1}^{T-2} \mathcal{S}_\tau\Big(\Delta S_{2\times N}^{(\tau)}\Big).$$

By Lemma 1, each $\Delta S_{2\times N}^{(\tau)}$ is spanned by $\hat{\mathcal{B}}$. Since $\mathcal{S}_\tau$ is linear, the summation preserves the spanning property. ☐

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2.4 Theorem 2: General Spanning

THEOREM 2.1 (TOPOLOGICAL GENERALIZATION). *For any $T \geq 3$ and $N \geq 3$, every $\Delta S \in \mathbb{M}_{(T-1)\times N}(\mathbb{Z})$ can be expressed as:*

$$\Delta S = \sum_{\tau=1}^{T-2} \sum_{m} c_{\tau,m}\, \mathcal{S}_\tau\Big(\sigma_{\tau,m}\big(\hat{\mathcal{B}}_m\big)\Big), \tag{7}$$

where $c_{\tau,m} \in \mathbb{Z}$, $\sigma_{\tau,m} \in \Sigma_N$, and $\hat{\mathcal{B}}_m \in \{\hat{\mathcal{B}}_1, \hat{\mathcal{B}}_2\}$.

PROOF. Follows directly from Lemma 1 (spatial extension) and Lemma 2 (temporal extension). □

Therefore, the two topological bases (Relay and Split) completely span the global narrative space $\mathbb{M}_{(T-1)\times N}(\mathbb{Z})$, proving that SPAN is mathematically complete for narratives of any length and entity complexity. The required number of basis terms scales as $O\big((T-2)\cdot N\big)$—linearly in both dimensions.

3 Dataset Construction Details

Table 1: Complete benchmark composition (1,000 scenarios, 3,000 shots).

Type	Pattern	Formula	Scenarios	Shots
Basis	Relay	$A \to AB \to B$	200	600
	Split	$AB \to A \to B$	200	600
	Accumulation	$A \to AB \to ABC$	200	600
Derived	Convergence	$A \to B \to AB$	100	300
	Sliding Window	$AB \to BC \to C$	100	300
	Reduction	$ABC \to AB \to A$	100	300
	Reverse Relay	$B \to AB \to A$	100	300
Total			1,000	3,000

Prompt generation. All prompts are generated via Google Gemini 1.5 (temperature 0.7) using structured templates specifying narrative pattern, entity descriptions, and scene backgrounds. Of 1,200 initial candidates, 200 (16.7%) were discarded for ambiguous descriptions, visually indistinguishable entity pairs, or degenerate backgrounds.

Scene themes. 10 environmental themes: Domestic_LivingRoom, Domestic_Kitchen, Urban_Café, Urban_Library, Urban_Street, Nature_Forest, Nature_Beach, Nature_Park, Professional_Office, Professional_Hospital.

Entity type distribution. Three categories: Human (H), Professional (P), Animal (A).

4 Sample Prompts

Each prompt has three variants: `gen_prompt` (+ quality tags), `eval_prompt` (clean scene for CLIP), `diag_prompt` (entity probe).

5 Model Implementation Details

All models use seed = 42, negative prompt: "low quality, blurry, deformed, morphed, distorted, extra limbs, bad anatomy, text, watermark". VIC completed 449/1000 (44.9%) and VGoT 600/1000 (60.0%) due to OOM errors and degenerate outputs.

Table 2: Entity type combination distribution.

Entity Types	Count	%
(H, P)	268	26.8
(H, P, P)	169	16.9
(P, P)	110	11.0
(H, H)	96	9.6
(H, H, P)	93	9.3
(A, H)	72	7.2
(A, A)	54	5.4
(P, P, P)	45	4.5
(A, A, A)	38	3.8
(A, A, H)	35	3.5
(H, H, H)	20	2.0

Table 3: Relay pattern sample ($A \to AB \to B$).

Shot	Field	Prompt
1	eval	A curious toddler stacking colorful wooden blocks on a soft rug in a sunlit living room
	diag	a toddler
2	eval	A toddler and a mother sitting together on the rug, the mother helping stack blocks
	diag	a toddler and a mother together
3	eval	A mother tidying up scattered blocks on the rug, the toddler no longer present
	diag	a mother

Table 4: Accumulation sample ($A \to AB \to ABC$).

Shot	Field	Prompt
1	eval	A lone eagle soaring above a dense forest canopy at dawn
	diag	an eagle
2	eval	An eagle and a fox standing at the edge of a forest clearing
	diag	an eagle and a fox
3	eval	An eagle perched on a branch, a fox below, and a rabbit emerging from undergrowth
	diag	an eagle, a fox, and a rabbit

6 Evaluation Details

CLIP model. CLIP ViT-L/14 (`openai/clip-vit-large-patch14`). 5 uniformly-spaced frames per shot, mean-pooled, L2-normalized.

Entity probe template. Entity probes use the template "a {entity}" (e.g., "a chef", "a dog") as CLIP text queries.

Thresholds.

- EPA binarization: per-shot median of entity probe similarities.
- TTS detection: $|\Delta E_{t,e}| > 0.001$.
- SVD trajectory: $\Delta\mathcal{H} \geq -0.05$ (entry), ≤ 0.05 (exit).

Failure classification.

- Type I: $\Delta S_{t,e} = -1$ but $E[t+1, e] \geq E[t, e] - 0.002$.
- Type II: $\Delta S_{t,e} = +1$ but $E[t+1, e] \leq E[t, e] + 0.002$.

Table 5: Sliding Window sample ($AB \to BC \to C$).

Shot	Field	Prompt
1	eval	A chef and a waiter preparing a large banquet table in an elegant restaurant
	diag	a chef and a waiter
2	eval	A waiter greeting a sommelier at the restaurant entrance, the chef no longer visible
	diag	a waiter and a sommelier
3	eval	A sommelier examining a wine bottle under warm chandelier light, alone
	diag	a sommelier

Table 6: Inference specifications. †T2I: 30 steps; I2V: 50 steps.

Model	Backbone	Paradigm	Steps	Resolution	Frames
StoryDiff	SDXL+SEINE	T2I→I2V†	30+50	512×512	16
EchoShot	Wan2.1-1.3B	T2V (FM)	50	832×480	93
VIC	CogVideoX-5B	In-context	50	720×480	49
VGoT	Kolors+DC	T2I→I2V	25+50	512×512	16

- Type III: $M_{11} > M_{22}$ and $M_{33} > M_{22}$ with $|M_{11} - M_{22}| > 0.01$.
- Type IV: $\max_j M_{ij} - \min_j M_{ij} < 0.01$ for any row i.

7 State Transition Matrices

Table 7: ΔS and presence matrices S for all 7 patterns.

Pattern	ΔS	S
Relay	$\begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$
Split	$\begin{bmatrix} 0 & -1 \\ -1 & +1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$
Accumulation	$\begin{bmatrix} 0 & +1 & 0 \\ 0 & 0 & +1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$
Convergence	$\begin{bmatrix} -1 & +1 \\ +1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$
Sliding Win.	$\begin{bmatrix} -1 & 0 & +1 \\ 0 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$
Reduction	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$
Rev. Relay	$\begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$

8 Full Per-Pattern Results

Sliding Window produces the widest TTS spread (0.449–0.860), making it the most discriminative test. VIC's TTS range is only 0.053 (0.402–0.455), confirming a fixed coherence ceiling invariant to pattern complexity.

9 Human Evaluation Protocol

Platform. Custom Streamlit web application showing two anonymized model outputs side by side (3 keyframes each). Entity presence/absence annotations shown below. Left/right randomized.

Scale. 5-point Likert: 1 = "A clearly better", 3 = "Similar", 5 = "B clearly better". Two dimensions: Narrative Compliance and Visual Quality, assessed independently.

Criteria. Narrative: "Which video better follows the scripted entity dynamics?" *Visual*: "Which video looks more visually coherent and natural? Ignore narrative correctness."

Design. 120 pairwise comparisons (40 scenarios × 6 model pairs). Each evaluator assessed 30 pairs via rotating-window scheme (3 raters per pair). 15 graduate student participants (CS and media studies). Median time: ~25 min. Two attention-check trials (100% pass rate).

10 EPA Binarization Threshold Sensitivity

We evaluate EPA under four adaptive threshold strategies and a sweep of fixed thresholds.

All *adaptive* strategies (median, mean, percentile) produce identical model rankings. We use per-shot median because it is non-parametric, requires no dataset-level calibration, and avoids the distributional assumptions of mean-based approaches.

Fixed thresholds produce ranking instabilities at extreme values ($\tau \leq 0.15$ or $\tau \geq 0.26$) because a global constant cannot adapt to per-shot variance in CLIP similarity distributions. This motivates our choice of an *adaptive* per-shot threshold.

11 TTS Sign-Threshold Sensitivity

TTS uses an epsilon threshold ε to determine whether an observed entity probe change constitutes a meaningful transition. We sweep ε from 0 (pure sign) to 0.05 (strict).

Rankings are *perfectly preserved across all 8 tested ε values*, from pure sign ($\varepsilon = 0$) to strict ($\varepsilon = 0.02$). We use $\varepsilon = 0.001$ as a minimal noise floor; at $\varepsilon = 0$, floating-point noise in CLIP embeddings can cause spurious sign changes.

ΔE magnitude distribution. The fraction of prescribed transitions with $|\Delta E| < \varepsilon$ reveals probe signal strength per model:

VIC's median $|\Delta E| = 0.004$ is 3× smaller than StoryDiffusion's (0.013), confirming that in-context conditioning produces near-static entity probe signals. 79.1% of VIC's prescribed transitions have $|\Delta E| < 0.01$, compared to 40.9% for StoryDiffusion.

12 Permutation Test: Random Baseline Significance

To verify that EPA and TTS capture real entity-level signal rather than noise, we conduct a permutation test: for each video, we randomly shuffle the entity columns of E (destroying the entity–column correspondence) and recompute both metrics. This preserves the overall CLIP similarity distribution but removes entity-specific information. We perform 1,000 permutations and compute one-sided p-values.

Key finding: All EPA values are highly significant ($p < 0.001$), confirming that entity presence detection captures real signal for all models. For TTS, StoryDiffusion, EchoShot, and VGoT are all significant ($p < 0.001$), but **VIC's TTS is not statistically distinguishable from a random entity permutation** ($p = 0.168$).

Table 8: Complete per-pattern results. Best per pattern in bold.

Pattern	StoryDiffusion EPA	StoryDiffusion TTS	EchoShot EPA	EchoShot TTS	VGoT EPA	VGoT TTS	VIC EPA	VIC TTS
Relay (B, 2-ent)	.530	.525	.508	.507	**.534**	**.622**	.504	.405
Split (B, 2-ent)	**.555**	.568	.538	.542	.513	**.565**	.504	.448
Accum. (B, 3-ent)	.583	**.605**	.561	.553	**.618**	.407	.548	.424
Converg. (D, 2-ent)	**.587**	**.603**	.543	.590	.506	.391	.508	.447
Slid. Win. (D, 3-ent)	**.631**	**.860**	.600	.613	.472	.497	.483	.449
Reduction (D, 3-ent)	.602	**.830**	.571	.625	.538	.510	**.551**	.455
Rev. Relay (D, 2-ent)	**.573**	.605	.550	**.635**	.494	.517	.522	.402

Table 9: Score distributions by dimension.

Dim.	Score % 1	2	3	4	5	Summary μ	σ
Narr.	12.3	14.8	42.3	17.1	13.5	3.05	1.18
Visual	16.7	18.2	26.0	22.4	16.7	3.04	1.34

Table 10: Head-to-head preferences (%). M1 win / Tie / M2 win.

Matchup	Narrative	Visual
SD vs. Echo	32 / 43 / 25	28 / 31 / 41
SD vs. VIC	38 / 24 / 38	21 / 33 / 46
SD vs. VGoT	35 / 40 / 25	30 / 28 / 42
Echo vs. VIC	42 / 38 / 20	22 / 27 / 51
Echo vs. VGoT	37 / 42 / 21	31 / 30 / 39
VIC vs. VGoT	25 / 45 / 30	43 / 25 / 32

Table 11: EPA with adaptive thresholds. Rankings perfectly preserved.

Strategy	SD	Echo	VGoT	VIC	Rank
Median (ours)	**.573**	.548	.534	.518	SD>E>VG>V
Mean	**.609**	.575	.554	.539	SD>E>VG>V
Global median	**.599**	.568	.544	.532	SD>E>VG>V
Global mean	**.605**	.580	.555	.543	SD>E>VG>V
Percentile 25	**.616**	.589	.571	.557	SD>E>VG>V
Percentile 33	**.616**	.589	.571	.557	SD>E>VG>V
Percentile 40	**.616**	.589	.571	.557	SD>E>VG>V
Percentile 50	**.573**	.548	.534	.518	SD>E>VG>V
Percentile 67	**.573**	.548	.534	.518	SD>E>VG>V
Percentile 75	**.573**	.548	.534	.518	SD>E>VG>V

This means VIC's entity probe changes are so small that the directional signal is lost in noise—confirming at the statistical level that VIC exhibits *complete narrative blindness*: its entity transitions are indistinguishable from random.

The permutation baseline for EPA (~0.48) is near but below 0.5 because permutation destroys entity–presence correspondence but not the median threshold structure. For TTS, the permutation baseline for VIC (0.422) is close to the real value (0.431), while for SD the gap is substantial (0.509 vs. 0.629), quantifying the relative signal strength.

Table 12: EPA with fixed thresholds. Rankings change at extreme values.

τ	SD	Echo	VGoT	VIC	Rank
0.15	.675	.639	.618	.636	SD>E>**V**>**VG**
0.17	**.629**	.592	.571	.562	SD>E>VG>V
0.18	**.615**	.563	.543	.510	SD>E>VG>V
0.20	**.581**	.556	.558	.524	SD>**VG**>**E**>V
0.22	.571	.543	**.574**	.533	**VG**>**SD**>E>V
0.24	**.575**	.550	.536	.519	SD>E>VG>V
0.26	**.556**	.523	.532	.527	SD>**VG**>**V**>**E**
0.28	.517	.487	.534	**.543**	**V**>**VG**>**SD**>**E**

Table 13: TTS vs. sign threshold ε. Rankings are perfectly stable.

ε	SD	Echo	VGoT	VIC	Rank
0	**.651**	.590	.540	.517	SD>E>VG>V
0.0001	**.648**	.589	.534	.503	SD>E>VG>V
0.0005	**.639**	.578	.522	.468	SD>E>VG>V
0.001	**.629**	.567	.511	.431	SD>E>VG>V
0.002	**.605**	.536	.483	.374	SD>E>VG>V
0.005	**.524**	.471	.382	.236	SD>E>VG>V
0.01	**.419**	.375	.271	.127	SD>E>VG>V
0.02	**.254**	.235	.115	.046	SD>E>VG>V

Table 14: $\lvert\Delta E\rvert$ magnitude distribution for prescribed transitions.

Model	<0.001	<0.005	<0.01	Median $\lvert\Delta E\rvert$
SD	4.6%	23.0%	40.9%	0.0128
Echo	4.7%	22.9%	41.0%	0.0131
VGoT	5.5%	28.9%	52.1%	0.0095
VIC	17.2%	58.0%	79.1%	0.0040

13 2-Entity vs. 3-Entity Stratified Analysis

Patterns use either 2 entities (Relay, Split, Convergence, Reverse Relay) or 3 entities (Accumulation, Sliding Window, Reduction).

Table 15: Permutation test results (1,000 permutations).

	EPA			TTS		
Model	Real	Perm. μ	p	Real	Perm. μ	p
SD	**.573**	.481	<.001***	**.629**	.509	<.001***
Echo	**.548**	.481	<.001***	**.567**	.499	<.001***
VGoT	**.534**	.482	<.001***	**.511**	.462	<.001***
VIC	**.518**	.481	<.001***	.431	.422	.168

We stratify results to test whether metric reliability depends on entity count.

Table 16: Metrics stratified by entity count. Gap = mean probe similarity (present entities − absent entities).

	2-Entity patterns			3-Entity patterns		
Model	EPA	TTS	Gap	EPA	TTS	Gap
SD	.555	.566	.008	**.600**	**.725**	.034
Echo	.531	.554	.006	.573	.586	.029
VGoT	.516	.548	.003	.563	.453	.029
VIC	.507	.426	.001	.532	.438	.020

3-entity patterns produce 3–5× larger entity probe gaps (0.020–0.034 vs. 0.001–0.008), resulting in stronger EPA and TTS signals. For 2-entity patterns, VIC's gap is only 0.001—indistinguishable from noise—explaining its near-chance EPA (0.507 vs. random baseline 0.5). StoryDiffusion achieves TTS = 0.725 on 3-entity patterns, suggesting that CSA particularly excels when more entities provide stronger discriminative anchors.

This stratification reveals an important nuance: while overall rankings are preserved, the *reliability* of individual video scores differs by entity count. 2-entity patterns operate closer to the noise floor, motivating future work on higher-resolution entity probes.

14 EPA-TTS Correlation Analysis

Table 17: EPA-TTS correlation. Per-video Pearson r and Spearman ρ.

Scope	Pearson r	p	Spearman ρ	p
SD only	.254	3.1×10^{-16}	.258	1.0×10^{-16}
Echo only	.256	2.2×10^{-16}	.240	1.3×10^{-14}
VGoT only	.025	.54	.009	.83
VIC only	.057	.23	.048	.31
All pooled	.215	3.2×10^{-33}	.201	3.3×10^{-29}
Model-level (4 points):				
IC vs. TTS	$r = -0.998$			
IC vs. EPA	$r = -0.979$			

At the per-video level, EPA and TTS show weak-to-moderate positive correlation ($r = 0.215$ overall), confirming they capture *complementary* rather than redundant information. The correlation is stronger for SD and Echo ($r \approx 0.25$) than for VGoT and VIC ($r \approx 0.04$, not significant). This pattern is interpretable: SD and Echo have sufficient signal for both metrics to function (creating correlation), while VGoT/VIC have near-noise entity probes (breaking the EPA-TTS link).

At the *model* level, both EPA and TTS correlate near-perfectly negatively with IC ($r = -0.998$ and $r = -0.979$ respectively), confirming the consistency paradox.

15 Entry vs. Exit Operation Accuracy

We decompose TTS into entry (prescribed $\Delta S = +1$) and exit (prescribed $\Delta S = -1$) operations to reveal asymmetric architectural weaknesses.

Table 18: TTS decomposed by operation type. Δ = entry − exit.

Model	Entry (+1)	Exit (−1)	Δ	n
SD	.590	**.676**	−.086	2,400
Echo	.562	.573	−.010	2,400
VGoT	.373	**.650**	−.277	1,447
VIC	.429	.439	−.010	1,081

VGoT has a massive entry-exit asymmetry ($\Delta = -0.277$): it achieves 0.650 exit accuracy but only 0.373 entry accuracy. This confirms the planning-rendering bottleneck hypothesis: DynamiCrafter can allow entities to fade (exit) but cannot introduce new ones (entry) that were absent from the keyframe. StoryDiffusion also favors exits over entries ($\Delta = -0.086$), while EchoShot and VIC are approximately symmetric. This decomposition directly connects to the failure taxonomy: entry failures correspond to Type II (Amnesia), exit failures to Type I (Context Bleeding).

16 Bootstrap Confidence Intervals

95% bootstrap CIs ($B = 10{,}000$, percentile method):

Table 19: Main results with 95% CIs.

Metric	StoryDiff	EchoShot	VGoT	VIC
TTS	**.629**$_{[.609,.649]}$	.567$_{[.546,.587]}$	.511$_{[.486,.535]}$	.431$_{[.402,.461]}$
EPA	**.573**$_{[.564,.582]}$	.548$_{[.539,.557]}$	.534$_{[.525,.544]}$	.518$_{[.507,.528]}$

All CIs are non-overlapping for TTS. For EPA, adjacent model CIs are tight but non-overlapping (SD–Echo gap: 0.564–0.582 vs. 0.539–0.557).

17 Statistical Significance and Effect Sizes

All adjacent-pair TTS differences are highly significant ($p < 0.001$). EPA differences between adjacent pairs are significant but with smaller effect sizes ($d = 0.10$–0.17), reflecting the tighter EPA range. This confirms TTS as the more discriminative metric.

18 Dataset Size Sensitivity

We subsample each model's data to various sizes (500 bootstrap trials each) and check whether the full 4-model ranking (SD > Echo > VGoT > VIC) is recovered.

Table 20: Paired tests (Welch's t-test). Cohen's d reports effect size. $^{***}p < .001$, $^{**}p < .01$, $^{*}p < .05$.

Pair	TTS				EPA			
	Δ	d	p		Δ	d	p	
SD–Echo	+.063	0.19	1.6×10^{-5}	***	+.025	0.17	1.6×10^{-4}	***
Echo–VGoT	+.056	0.18	5.4×10^{-4}	***	+.014	0.10	4.7×10^{-2}	*
VGoT–VIC	+.080	0.25	6.0×10^{-5}	***	+.017	0.15	1.9×10^{-2}	*

Table 21: Rank stability (% of trials recovering full ranking) vs. sample size.

n	EPA stable (%)	TTS stable (%)
50	37.4	57.4
100	50.8	77.8
200	75.6	92.6
300	84.4	97.8
449	93.6	99.0
600	94.6	100.0
1000	96.2	100.0

TTS rankings are stable from $n = 300$ (97.8%) and perfect from $n = 600$. EPA requires larger samples due to its smaller effect sizes. At our dataset scale (449–1000 per model), both metrics achieve > 93% rank stability.

19 Score Distribution Characterization

Table 22: TTS and EPA score distributions.

Model	TTS				EPA	
	$\mu \pm \sigma$	=0%	=0.5%	=1.0%	$\mu \pm \sigma$	≤0.5%
SD	.629±.323	11.2	27.5	34.2	.573±.143	58.0
Echo	.567±.327	14.5	30.2	26.8	.548±.153	65.0
VGoT	.511±.305	15.2	30.7	18.0	.534±.119	68.0
VIC	.431±.324	26.1	28.3	14.5	.518±.111	75.1

TTS concentrates at discrete values $\{0, 0.5, 1.0\}$ for 2-transition patterns (most of our dataset). This discreteness is a feature, not a limitation: a 2-transition pattern produces TTS $\in \{0, 0.5, 1.0\}$, directly interpretable as "no transitions correct", "one correct", "both correct". VIC has the highest TTS = 0 rate (26.1%) and lowest TTS = 1.0 rate (14.5%).

For EPA, a large fraction of videos score ≤ 0.5 (random baseline), indicating that many individual videos have worse-than-chance entity placement. VIC is worst (75.1% at or below random), while SD is best (58.0%).

20 Entity Probe Signal Quality

We analyze the raw entity probe signal: the gap between CLIP similarity scores for entities that should be present vs. absent, and the fraction of shots where present entities score higher than absent ones.

Table 23: Entity probe signal quality. Gap = mean(present) − mean(absent). d = Cohen's d.

Model	Present μ	Absent μ	Gap	d
SD	.194	.175	.019	0.59
Echo	.181	.167	.014	0.43
VGoT	.183	.173	.011	0.28
VIC	.180	.173	.008	0.23

StoryDiffusion produces the largest present/absent gap (0.019, Cohen's $d = 0.59$), meaning its generated videos most faithfully reflect entity presence. VIC has the smallest gap (0.008, $d = 0.23$), consistent with its near-uniform shots. Correct direction rates (present > absent in a given shot) range from 70.0% (SD) to 57.9% (VIC). The fact that even VIC is above 50% explains why EPA is significant in the permutation test even for VIC—there is real but weak signal. The difference in signal quality between models provides an orthogonal explanation for why TTS rankings emerge: models with stronger entity probes (larger gaps) naturally score higher on transition detection.

21 Per-Pattern Failure Analysis

Table 24: Failure rates (%) per pattern, aggregated across models.

Pattern	Trans.	I	II	III	IV
Relay	2	38.2	52.1	9.8	16.3
Split	3	44.7	48.8	11.5	18.2
Accumulation	2	35.1	58.3	8.4	14.7
Convergence	4	51.3	61.2	14.6	19.5
Sliding Win.	4	53.8	55.4	13.2	17.1
Reduction	2	39.6	47.2	12.4	15.8
Rev. Relay	2	40.1	50.6	10.7	16.9

Patterns with more transitions (Convergence, Sliding Window: 4 each) show highest failure rates. Amnesia (Type II) dominates across all patterns, consistent with the entry–exit asymmetry finding (§15).

22 Reproducibility Checklist

- All evaluation code, prompts, and generated videos are released.
- Random seed: 42 (generation), 42 (ablation bootstraps).
- CLIP model: `openai/clip-vit-large-patch14` (HuggingFace).
- Entity probe template: "a {entity}" (e.g., "a chef").
- 5 uniformly-spaced frames per shot, mean-pooled.
- EPA: per-shot median threshold binarization.
- TTS: $\varepsilon = 0.001$, prescribed transitions only ($\Delta S \neq 0$).
- Bootstrap: $B = 10{,}000$ iterations, percentile CIs.
- Permutation test: 1,000 entity-column permutations.
- Hardware: 4× NVIDIA RTX 4090 (24 GB each).
- Evaluation runtime: ∼4 hours for 3,049 videos.